

CS 103000

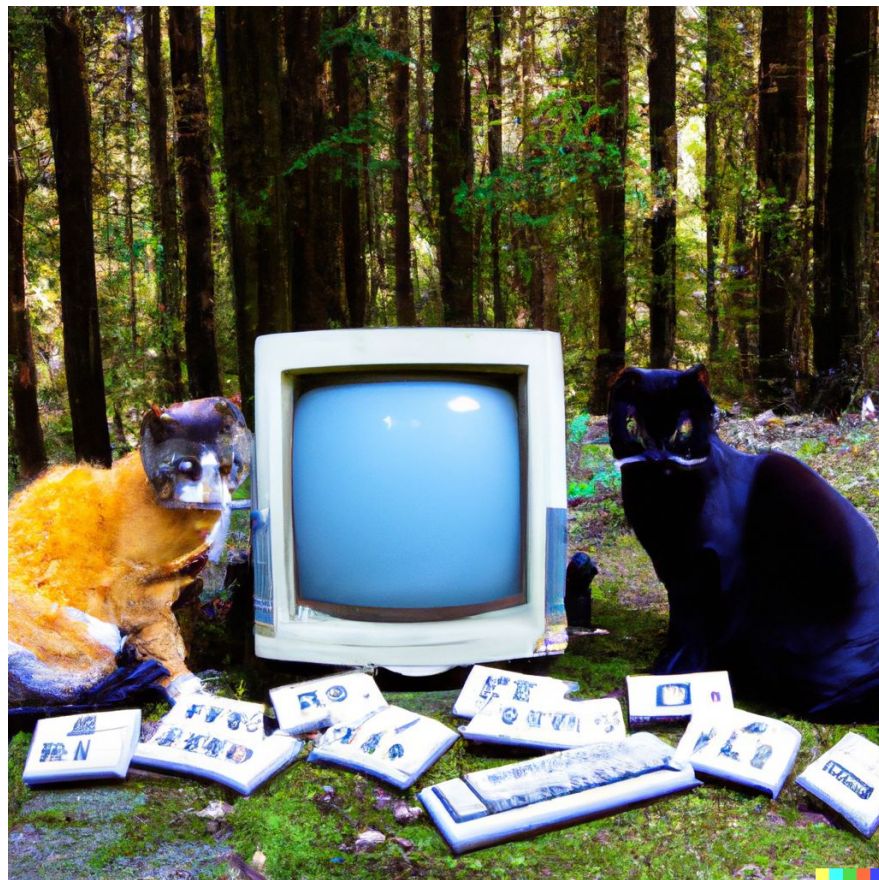
Prof. Madeline Blount

Week 5:

LOOPS (part 2)



NESTED LOOPS

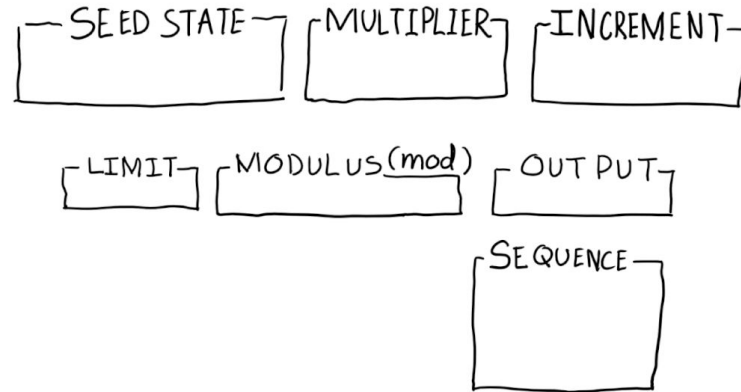


Dall-E 2: cats learning C++ in the forest on '90's technology

- $x = ((a * x) + c) \% m$
- if: $a \text{ (multiplier)} = 3$
- if: $c \text{ (incrementor)} = 4$
- if: $m \text{ (modulo \%)} = 15$
- if: $x_0 = 1$
- Write out the first 6 numbers of this sequence.

Pseudorandom! a **simulation**, random enough

- `rand()` = Linear Congruential Generator (LCG)
- $x = ((a * x) + c) \% m$
- Next number based on the previous (**state**)



for loops!

```
int i;  
for (i = 0; i < max; i++) {  
    // loop in here  
}
```



Special Pythagorean triplet

Problem 9



A Pythagorean triplet is a set of three natural numbers, $a < b < c$, for which,

$$a^2 + b^2 = c^2$$

For example, $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

There exists exactly one Pythagorean triplet for which $a + b + c = 1000$.
Find the product abc .

euler #9 answer: 31875000

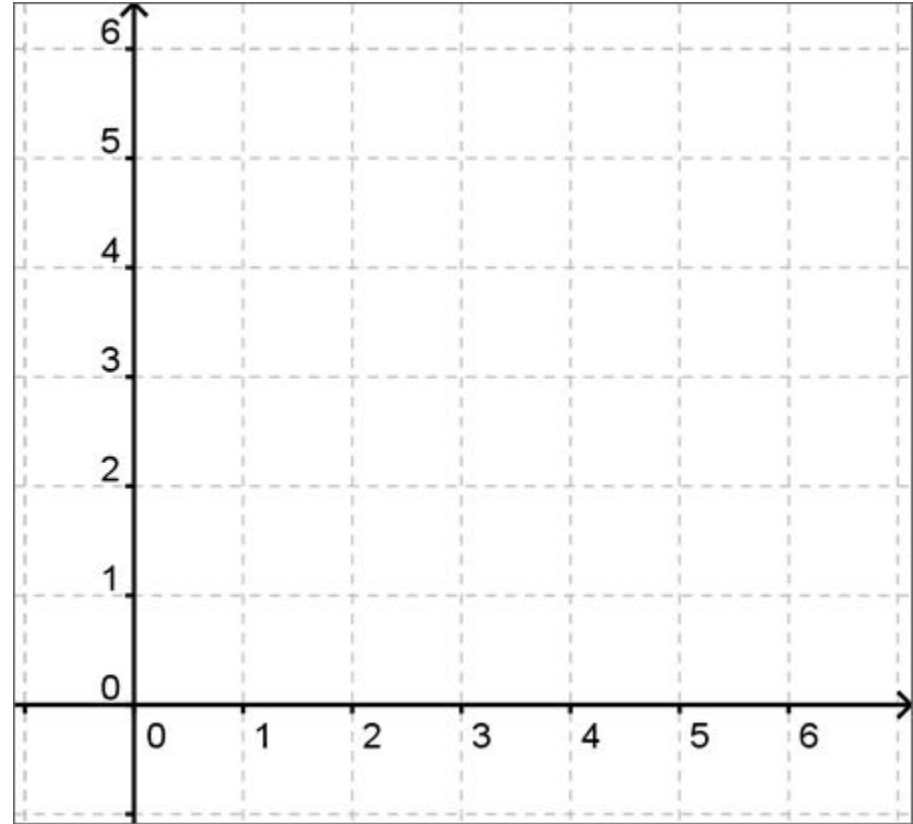
"brute force"



using loops + computation power

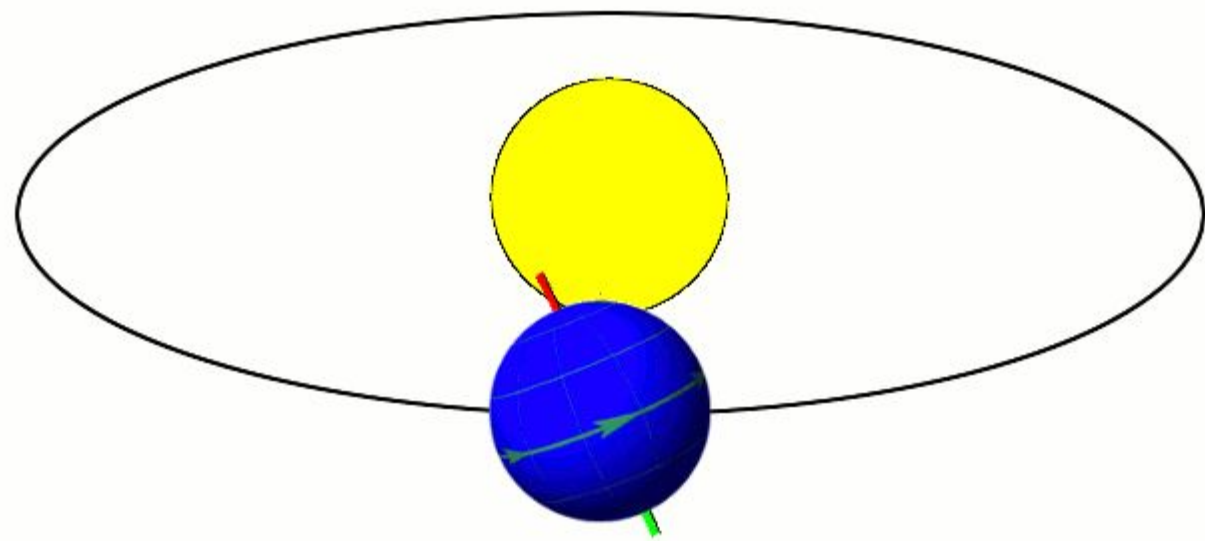


Nested for loops:
MULTI-DIMENSIONAL







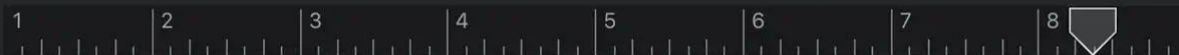




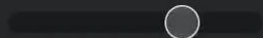
My Songs



FX



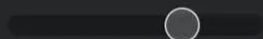
Hip Hop Drum Machine



Hip Hop Drum Machine



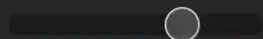
Picked



Picked



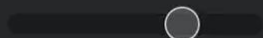
Classic Clean



Classic Clean



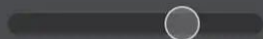
Ethereal Rhythm



Ethereal Rhythm



Hard Rock



Hard Rock





Smallest Multiple

Problem 5



2520 is the smallest number that can be divided by each of the numbers from 1 to 10 without any remainder.

What is the smallest positive number that is evenly divisible by all of the numbers from 1 to 20?